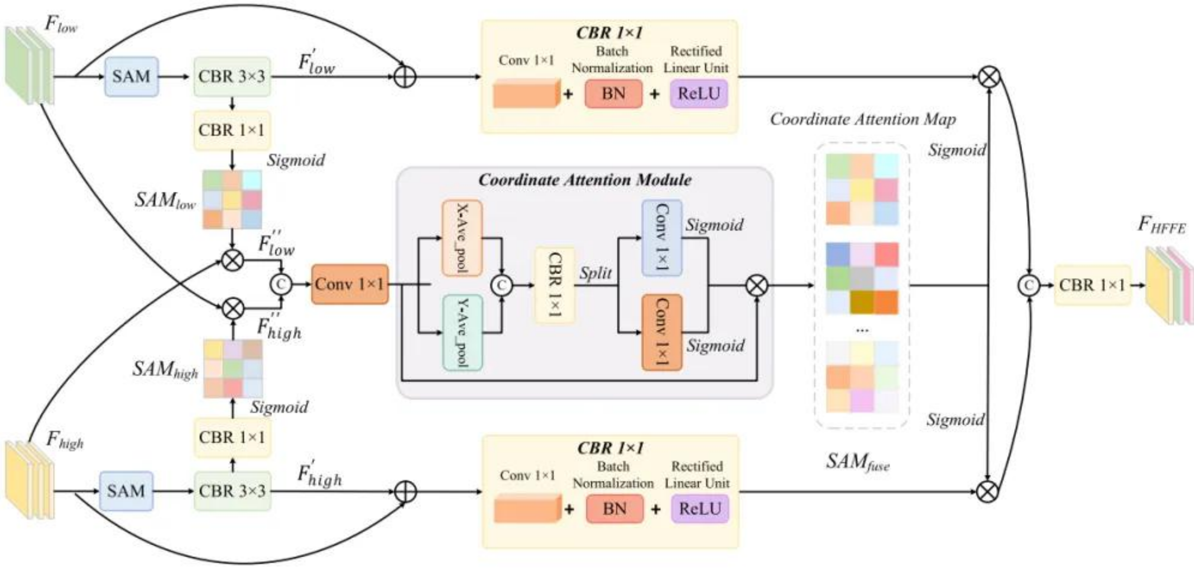
核心速览



论文题目： HAFNet: Hierarchical Attention Fusion Network for Infrared Small Target Detection

中文题目： HAFNet：红外小目标检测的层次注意力融合网络

论文链接：

https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=11154006

所属单位： 江西财经大学等

核心速览： 本文提出了一种基于 U-Net 架构的新型分层注意力融合网络（HAFNet），用于红外小目标检测（IRSTD），通过设计双分支语义感知模块和分层特征融合编解码器，显著提升了复杂场景中小目标的检测性能。

论文内容详细解读：https://mp.weixin.qq.com/s/VGkQBBC_P3C1VE5HCU-bZg

```
    )
    (conv3): CBR(
      (conv): Conv2d(64, 32, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (bn): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      (act): ReLU()
    )
    (Up_to_2): Upsample(scale_factor=2.0, mode='bilinear')
    (feature_low_sa): SpatialAttention(
      (conv1): Conv2d(2, 1, kernel_size=(7, 7), stride=(1, 1), padding=(3, 3), bias=False)
      (sigmoid): Sigmoid()
    )
    (feature_high_sa): SpatialAttention(
      (conv1): Conv2d(2, 1, kernel_size=(7, 7), stride=(1, 1), padding=(3, 3), bias=False)
      (sigmoid): Sigmoid()
    )
    (ca): CoordAttiton(
      (pool_h): AdaptiveAvgPool2d(output_size=(None, 1))
      (pool_w): AdaptiveAvgPool2d(output_size=(1, None))
      (conv1): Conv2d(32, 8, kernel_size=(1, 1), stride=(1, 1))
      (bn1): BatchNorm2d(8, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      (act): h_swish(
        (sigmoid): h_sigmoid(
          (relu): ReLU6(inplace=True)
        )
      )
      (conv_h): Conv2d(8, 32, kernel_size=(1, 1), stride=(1, 1))
      (conv_w): Conv2d(8, 32, kernel_size=(1, 1), stride=(1, 1))
    )
    (conv_final): CBR(
      (conv): Conv2d(64, 32, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (bn): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      (act): ReLU()
    )
  )

哔哩哔哩/微信公众号: AI缝合术, 独家整理!

Input: torch.Size([1, 32, 128, 128]) torch.Size([1, 32, 128, 128])
Output: torch.Size([1, 32, 128, 128])
```